Efficient Exploration by Novelty Pursuit

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International Conference on Distributed Artificial Intelligence (DAI), 2020

October 12, 2020

Overview

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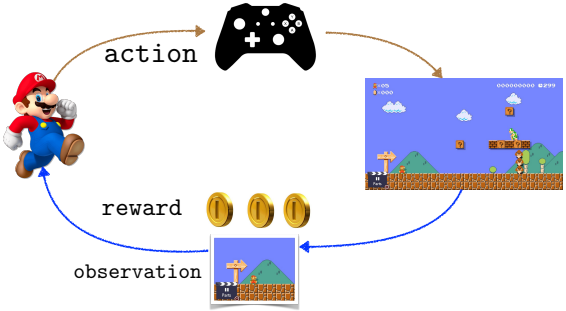
Reinforcement Learning

- RL is a learning paradigm that an agent interacts with the unknown environment to find the optimal decisions.



- RL directly learns from stochastic feedbacks (s, a, r, s').

Exploration v.s. Exploitation

- In an unknown environment, the agent is uncertain about the possible outcomes.

- `Exploration`: investigate the unknown actions which may bring large returns or unexpected losses.
- `Exploitation`: implement the well-known but possibly sub-optimal actions.

Towards Efficient Exploration

- Simple exploration strategies based on "dithering" methods are inefficient.
  - ϵ-greedy and Boltzmann strategy require almost $\mathcal{O}(2^N)$ samples to make progress on deep-sea. [Osband et al., 2019].

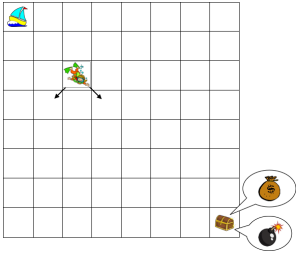


Figure 2: Deep-sea. Figure from [Osband et al., 2019]. There are only two actions and the reward is released at the most bottom right corner.

Towards Efficient Exploration

- Theoretically, efficient exploration requires to "write-off" the known and inferior actions.
- There are general two principles borrowed from bandits: optimism in the face of uncertainty (`OFU`) and Thompson sampling (`TS`).
  - `OFU`: add "reward bonus" by constructing upper confidence intervals [Stadie et al., 2015, Pathak et al., 2017, Burda et al., 2019b].
  - `TS`: sample the plausible actions from the iteratively updated posterior distribution [Osband et al., 2016a,b, O'Donoghue et al., 2018].

Efficient Exploration for Large-scale MDPs

- Large-scale MDPs (e.g., `Montezuma's Revenge`, `SuperMarioBros`, and `StarCraft`) are very challenging.
  - State space (and action space) is huge.
  - Planning horizon is long.
  - Rewards are sparse.
- `OFU` may lead to unexpected results and the weights to balance the tradeoff cannot be learnt from environments [Fortunato et al., 2018, Burda et al., 2019a].
- It's intractable to compute the posterior distribution to perform `TS` [Osband et al., 2019, Russo et al., 2018].

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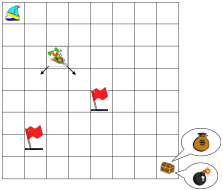
Novelty-Pursuit: Intuition

- Large-scale MDPs require to efficiently visit the whole state space.
- Goal-conditioned policy is proficient in reaching the desired state.



- We can learn a (near)-optimal policy from the diverse experience generated by a class of goal-conditioned policies that explore the whole state space.

Novelty-Pursit: Method

- (Exploration) We mark seldomly visited states as the targets for the goal-conditioned policy.
- (Exploration) Once the target is achieved, the agent performs random actions to discover new states.
- (Exploitation) A deployment (evaluation) policy learns from the collected experience by off-policy learning.

Novelty-Pursit: Method

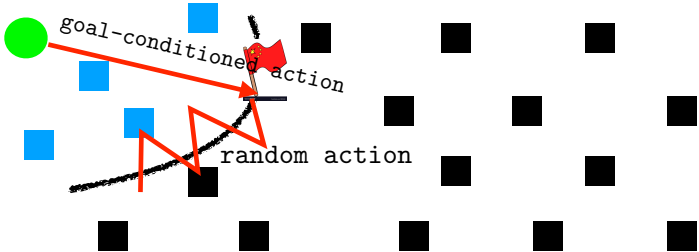


Figure 3: Illustration for novelty-pursuit. First, a goal-conditioned policy plans to reach the exploration boundary; then, it performs random actions to discover new states.

Novelty-Pursit: Method

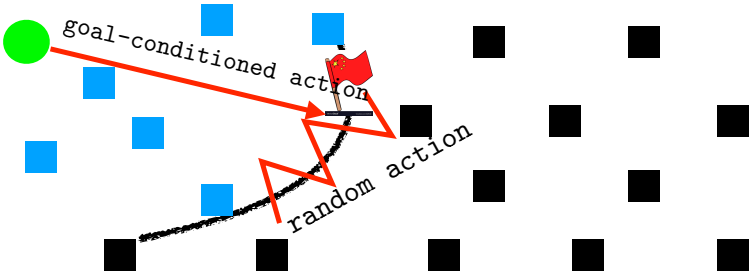


Figure 4: Illustration for novelty-pursuit. First, a goal-conditioned policy plans to reach the exploration boundary; then, it performs random actions to discover new states.

Novelty-Pursuit: Select Goals

- We select novel states from the buffer $\mathcal{B}$ to set goals.
- There are many approximation methods to measure novelty.
  - To name a few, pseudo-count [Bellemare et al., 2016, Ostrovski et al., 2017], prediction error [Stadie et al., 2015, Pathak et al., 2017], random network distillation (RND) [Burda et al., 2019b], etc.
- In practice, we maintain a small priority buffer to store the recent novel states.

Novelty-Pursuit: Train Goal Policy

- Training the goal-conditioned policy with sparse rewards (e.g., 1 for success and 0 for failure) is difficult.
- We use hindsight experience replay (`HER`) [Andrychowicz et al., 2017] (or `reward shaping`) to accelerate learning.
  - `HER`: replace each episode with an achieved goal rather than one that the agent was trying to achieve.
  - `Reward shaping`: introduce additional training rewards with mild conditions to guide the agent.

Novelty-Pursuit: Train Deployment Policy

- We use off-policy learning algorithms to learn the deployment (evaluation) policy from the experience buffer.
  - ACER [Wang et al., 2017] for environments with the discrete action space.
  - DDPG [Lillicrap et al., 2016] for environments with the continuous action space.
- To remedy the issue of extrapolation error [Fujimoto et al., 2019] when doing off-policy learning, we allow this policy to interact with environments occasionally.

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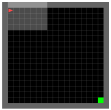
E1: The Efficiency of Exploration

- We use the entropy of visited states distribution to measure the efficiency of exploration.
  - A larger entropy implies a more flat distribution and thus is better.
- We consider the baselines: `random` walk and the exploration method based on reward `bonus` [Burda et al., 2019b].
- We also consider the variants of our method with oracles.
  - `planning-oracle`: the oracle provides the shortest path to reach goals.
  - `counts-oracle`: the oracle provides exact counts to select the seldomly visited states.

E1: The Efficiency of Exploration

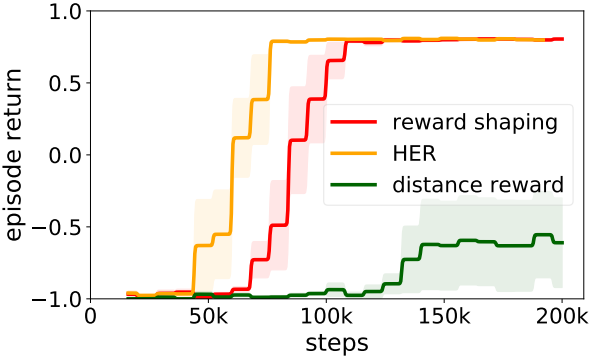
Table 1: Average entropy of visited state distribution over 5 random seeds on the 17×17 maze environment.

	Entropy
random	5.129 ± 0.021
bonus	5.138 ± 0.085
novelty-pursuit	$\mathbf{5.285 \pm 0.073}$
novelty-pursuit-planning-oracle	5.513 ± 0.077
novelty-pursuit-counts-oracle	5.409 ± 0.059
novelty-pursuit-oracles	5.627 ± 0.001
maximum	5.666



E2: Training of Goal Policy

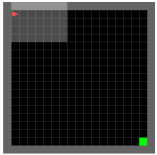
- We test the technique of goal-conditioned policy training on the 17×17 maze environment.

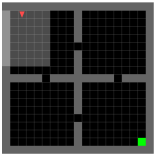


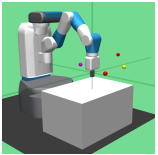
- `HER` and `reward shaping` work well.

E3: Performance on Complicated Tasks

- Baselines: `bonus` [Burda et al., 2019b] and `vanilla` method described below.
  - ACER [Wang et al., 2017] (with entropy regularization): Empty Room, Four Rooms and SupperMarioBros.
  - DDPG [Lillicrap et al., 2016] (with Gaussian noise): Fetch Reach [Andrychowicz et al., 2017].
- Environments: `Empty Room`, `Four Rooms`, `Fetch Reach` and `SuperMarioBros`.







E3: Performance on Complicated Tasks

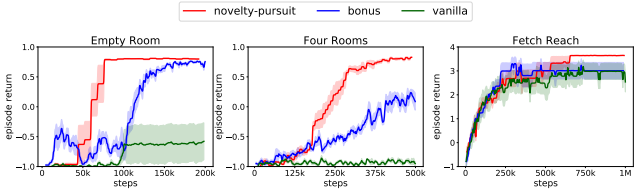


Figure 5: Training curves of learned policies over 5 random seeds on the Empty Room, Four Rooms, and FetchReach environments.

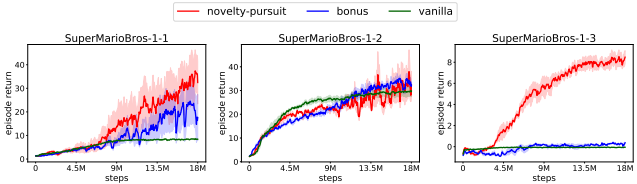


Figure 6: Training curves of learned policies over 3 random seeds on the game of SuperMarioBros.

E3: Performance on Complicated Tasks

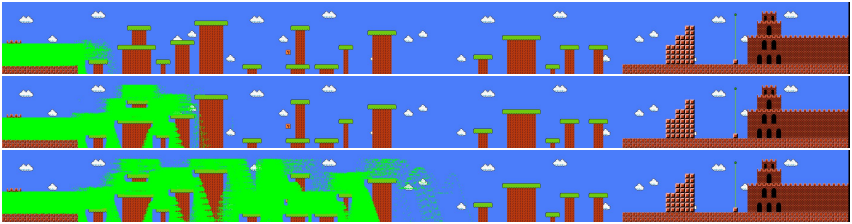


Figure 7: Trajectory visualization on SuperMarioBros-1-3. Trajectories are plotted in green cycles with the same training samples (18 million). The agent starts from the most left part and needs to fetch the flag on the most right part. Top row: vanilla ACER; middle row: ACER + exploration bonus; bottom row: novelty-pursuit (ours).

Video is available at http://www.liziniu.org/images/mario-video.mp4.

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- The proposed method novelty-pursuit is efficient to handle some large-scale MDPs like SuperMarioBros.
  - First, a goal-conditioned policy plans to reach the exploration boundary;
  - Subsequently, it performs random actions to discover new states.
- We will consider to extend this idea into solving more challenging tasks like `Montezuma's Revenge` and `Sonic the Hedgehog`.

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Acknowledgements

We thank Fabio Pardo for sharing ideas to visualize trajectories for SuperMarioBros. In addition, we appreciate the helpful instruction from Dr. Yang Yu and the insightful discussion with Tian Xu as well as Xianghan Kong.